VectorSonic AI Problem Statement

Despite significant advances in artificial intelligence and robotics, current mosquito control methods fail to provide a fully autonomous, end-to-end solution for both disease vector detection and elimination. The existing approaches are fundamentally fractured: detection systems are largely passive (relying on manual traps or data-logging microphones), while elimination methods rely on broad, environmentally harmful chemical fogging. Furthermore, attempts to automate this process using computer vision consistently fail because disease vectors are tiny, fast-moving, and primarily active in the dark.

Currently, there is a critical lack of closed-loop, autonomous systems capable of real-time, species-specific acoustic detection (such as MFCC-based analysis) directly linked to targeted, localized elimination. Without a system that can accurately 'hear' and instantly neutralize a threat without human intervention, public health efforts will continue to suffer from delayed outbreak responses, high manual labor costs, and severe ecological damage.